

The climate impact of improved cookstove subsidies

Susanna B. Berkouwer¹ Joshua T. Dean²
(they/them) (he/him)

¹The Wharton School, University of Pennsylvania and NBER

²The Booth School of Business, University of Chicago

September 2026

Kenyan Jiko and Jikokoa cookstoves

Kenyan Jiko
\$2-\$5



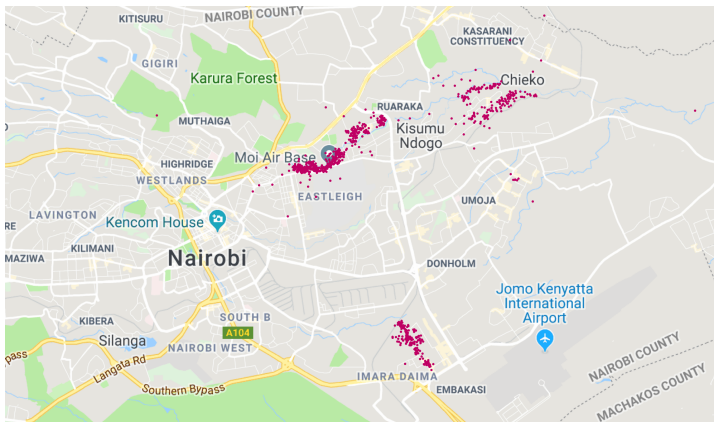
BURN Jikokoa Stove
\$40 (subsidized: \$10)



- ▶ Identical inputs: same charcoal
- ▶ Identical usage: almost no learning or behavior change
- ▶ Improved insulation materials (e.g. ceramic wool)
- ▶ Produced at factory outside Nairobi (Ruiru)

Randomized trial with 955 households in Nairobi, Kenya

- ▶ Field officers knock on doors to invite daily charcoal users to participate
- ▶ 95% of respondents are women
- ▶ Households have low incomes: USD 3–9 per day
- ▶ Charcoal spending is high: USD 0.5–1 per day



Converting charcoal usage to CO₂e emissions

(1) Charcoal spending to KG charcoal purchased

- ▶ Annual household charcoal spending: \$290
- ▶ At local market price (\$0.30/KG): 969 KG charcoal per year

(2) KG charcoal purchased to tons of CO₂e emitted

- ▶ Net calorific value: 0.027 TJ per tonne of charcoal
- ▶ Emissions factor: 162 tCO₂e per TJ

Gill-Wiehl, Kammen, Haya, 2024; Floess et al 2023; Whitman, Lehmann 2011

- ▶ 4:1 firewood-to-charcoal conversion ratio
- ▶ 38% fNRB
- ▶ Combustion (38) and non-combustion (124) emissions
- ▶ Each KG of charcoal burnt in urban Nairobi emits 4.4 KG CO₂e

Today: Synthesizing three academic articles

1. "Credit, attention, and externalities in the adoption of energy efficient technologies by low-income households." **American Economic Review** (2022).
2. "Cooking, health, and daily exposure to pollution spikes." **American Economic Journal: Economic Policy** (2026).
3. "The cost-effectiveness of clean cookstove carbon mitigation after adjusting for additionality and impact." **Climate Change Economics** (2026).

Research papers and policy summaries are available at:

www.susannaberkouwer.com/research

Funding and ethics approvals

Funding:

The Weiss Fund

for Research in Development Economics



Ethics approvals:

- ▶ Kenya: KEMRI/RES/7/3/1; AMREF ESRC P1195/2022
- ▶ United States: UC Berkeley CPHS 2017-11-2534; University of Pennsylvania 849048; University of Chicago IRB22-0943.

Today

Introduction

Global climate markets

Additionality

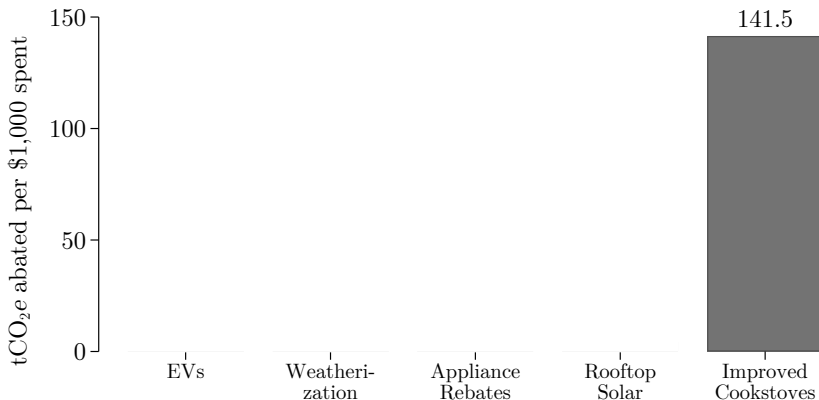
Impact

Climate and additionality

The global cooking sector

- ▶ One charcoal stove emits approx. **4.2 tCO₂e** per year
- ▶ For comparison: driving a gasoline vehicle emits **4.6 tCO₂** per year
 - ▶ EPA (2025): Average U.S. household drives 11,500 miles at 22.2 MPG

How many tons of CO₂e do green tech subsidies abate?



R. W. Hahn, N. Hendren, R. D. Metcalfe, and B. Sprung-Keyser. "A welfare analysis of policies impacting climate change." **American Economic Review** (2026).

How do we design effective carbon markets?

1. Buyers must differentiate between products being sold
 - ▶ A patient wants to know which medications work (and which do not)
 - ▶ The average efficacy of a group of medications does not aid the patient
2. Independent evaluators should generate objective information
 - ▶ In the U.S. the FDA regulates the testing of new medications
3. Generating objective information is costly
 - ▶ Randomized trials and independent data collection are expensive
4. Those costs are worth a well-functioning market
 - ▶ Life expectancy goes up when patients can access effective medications

Unified global carbon market proposal

Authors: Rohini Pande, Robin Burgess, Maryam Farboodi, Agnes Norris Keiller, Lucy Page (2025)

- ▶ *Pillar 1*: an opt-in compliance market with a global emissions cap that declines over time. Jurisdictions join voluntarily but then commit to common rules; finance flows to low- and middle-income countries.
- ▶ *Pillar 2*: MARVIN, a Measurement, Accounting, Risk management, and Verification Institution to ensure credit quality.

More information:

egc.yale.edu/carbon-markets



Efficiency of improved cookstove subsidies

$$\text{Subsidy efficiency} = \frac{\text{average tons of CO}_2\text{e abated}}{\text{dollar expenditure}}$$

Two central concerns:

Term	Question	Risks
Additionality	How much do subsidies increase the number of improved stoves sold?	Non-additional spending
Impact	How much does a subsidized stove sale reduce emissions?	Stacking, rebound, low usage, breakage, control group adoption

$$\text{Subsidy efficiency} = \underbrace{\frac{\text{stove sales}}{\text{dollar expenditure}}}_{\text{Additionality}} \cdot \underbrace{\frac{\text{average tons of CO}_2\text{e abated}}{\text{stove sale}}}_{\text{Impact}}$$

Today

Introduction

Global climate markets

Additionality

Impact

Climate and additionality

Each respondent was randomly assigned a subsidy amount

- ▶ High prices: \$26 (35% discount)
- ▶ Low prices: \$11 (73% discount)

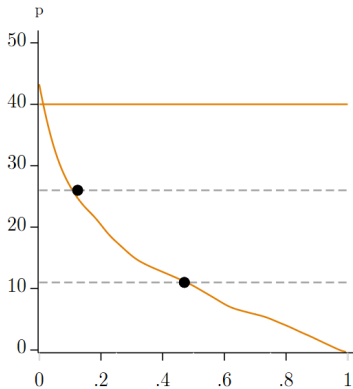


Why randomize?

- ▶ No differences in economic, social, or demographic household characteristics
- ▶ Any differences in outcomes are 100% due to the subsidy difference

Each respondent could buy at the price they were offered

- ▶ At the high price: 8% chose to buy
- ▶ At the low price: 48% chose to buy



Giving 100 individuals a low or high subsidy

Low subsidy

Number of stoves sold	7.6
Total cost (additional buyers)	
Total cost (non-additional buyers)	
Total cost	\$107

Additional cost
Additional stoves sold
Subsidy cost per additional stove sold

- ▶ 16% of subsidy spending flows to non-additional buyers: t wanted to buy even with a low subsidy
- ▶ Every \$1,000 in subsidy spending generates 31 additional

Today

Introduction

Global climate markets

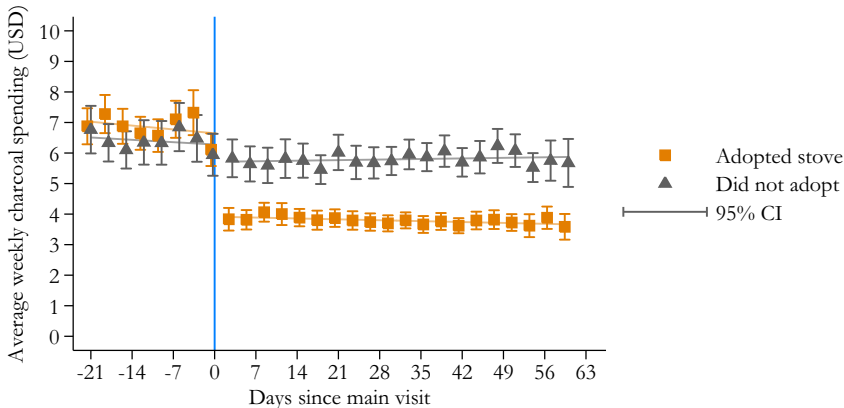
Additionality

Impact

Climate and additionality

Stove reduces charcoal use by \$2.19 per week (39%)

- ▶ Every respondent gets an SMS asking about charcoal spending every 3 days for 2 months

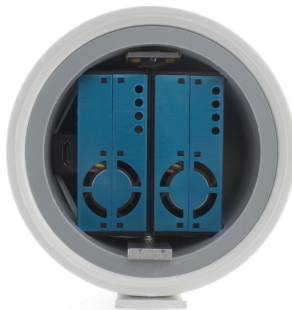


We weigh KG ash output from 1 month of charcoal use



3.5 years after purchase (2022-2023): Purple Air devices

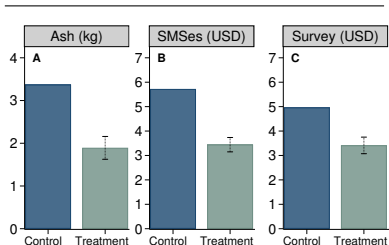
- ▶ Devices measure PM_{1.0} and PM_{2.5} (particles under $< 1.0\mu m$, $< 2.5\mu m$)
- ▶ One reading every 2 minutes
- ▶ Calibrated using Ward et al. (AGU 2021)
- ▶ Methodology and design developed by Berkeley Air Johnson et al 2021
- ▶ Best practices Chillrud et al 2021; Burrowes 2019; Gordon et al 2014; Gould et al 2022
- ▶ Dropped off during visit 1, picked up ~48 hours later (backpack has a battery)



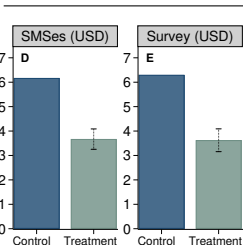
Improved stoves reduce charcoal usage 27–48%

- ▶ The independent variable is *ownership*: the presence of rebound effects, low usage, has lowered these estimates from what they would otherwise be

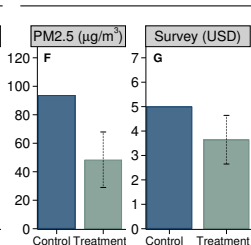
1 month later



1 year later



3.5 years later



- ▶ Weight of ash generated by charcoal usage
- ▶ SMS surveys
- ▶ In-person surveys
- ▶ Phone survey
- ▶ Pollution measurement

Impact: Ownership abates 1.7 tCO₂e per year

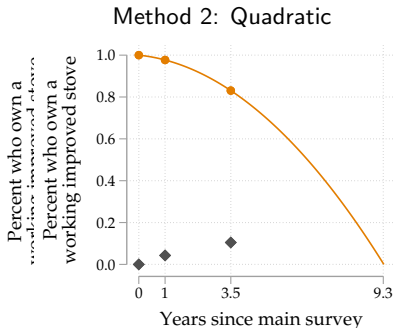
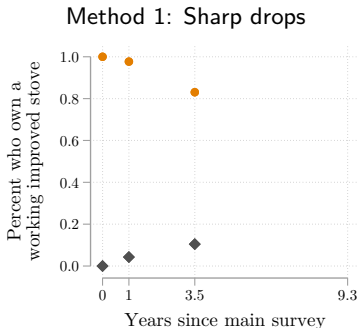
- ▶ Ownership reduces charcoal usage 27–48% across 7 independent outcomes
- ▶ Mean estimate: 40%
- ▶ Annual charcoal spending reduction of \$114
- ▶ Annual charcoal usage reduction of 379 kg
- ▶ 1 kg charcoal $\Rightarrow$ 4.4 kg CO₂e

Annual abatement of 1.7 tCO₂e per year per owner

Cookstove ownership over time

	Did not buy stove from us	Bought stove from us
<i>One month later</i>		
Owns Jikokoa	0%	98%
<i>One year later</i>		
Owns Jikokoa	5%	98%
<i>Three years later</i>		
Owns Jikokoa	10%	84%
Owns other wood/charcoal stove	88%	34%
Owns LPG stove	57%	62%
Owns bio-ethanol stove	15%	16%
Owns electric stove	0%	2%

How many additional years of ownership did the subsidies generate?



1. **Total lifetime:** Each stove lasts on average 3.1 to 6 years (we use 3.1)
2. **Additional duration:**
 - ▶ Control group purchases generate 0.31 stove years
 - ▶ Subsidized sales generate an *additional* 2.8—5.4 stove years (we use 2.8)

Today

Introduction

Global climate markets

Additionality

Impact

Climate and additionality

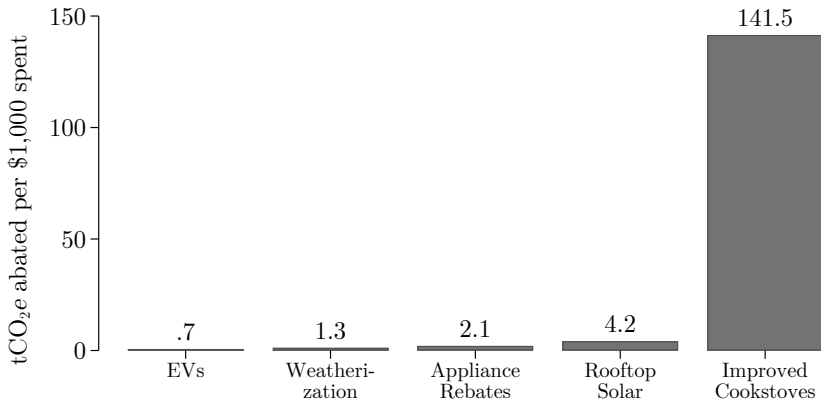
Aggregate climate impact of improved cookstove subsidies

- ▶ Every \$1,000 in subsidy spending generates 31 additional purchases
- ▶ Each sale generates 2.8 additional years of working stove ownership
- ▶ Each year of ownership abates 1.7 tCO₂e

Every \$1,000 in subsidy spending abates 144 tons of CO₂e

Scenario	Abatement per \$1,000
Paper estimate (sharp breakage; 38% fNRB; 4:1 firewood-to-charcoal)	144 tCO ₂ e
Quadratic breakage	282 tCO ₂ e
fNRB of 10%	124 tCO ₂ e
fNRB of 100%	187 tCO ₂ e
6:1 firewood-to-charcoal conversion ratio	195 tCO ₂ e

How many tons of CO₂e do green tech subsidies abate?



Estimates for the four other technologies:

R. W. Hahn, N. Hendren, R. D. Metcalfe, and B. Sprung-Keyser. “A welfare analysis of policies impacting climate change.” **American Economic Review** (2026).

Carbon mitigation: How to move forward?

Lessons:

1. There exist very high-quality improved cookstove projects
2. Cookstove projects can be imperfectly additional, experience some breakage rate, and have other issues, and still be highly effective
3. High-quality cookstoves are among the most effective large-scale climate mitigation technologies in the world
4. High-quality cookstoves are available today

Action items:

1. State-of-the-art technologies can measure usage, switching, emissions
2. **Challenge:** Quantifying additionality and impact requires a rigorous research strategy: subsidy recipients *relative to* non-recipients
3. **Opportunity:** Any project that can randomize prices and re-survey buyers can quantify additionality and impact
4. Credible, independent measurement can identify high-quality projects

More information: Three academic articles

1. "Credit, attention, and externalities in the adoption of energy efficient technologies by low-income households." **American Economic Review** (2022).
2. "Cooking, health, and daily exposure to pollution spikes." **American Economic Journal: Economic Policy** (2026).
3. "The cost-effectiveness of clean cookstove carbon mitigation after adjusting for additionality and impact." **Climate Change Economics** (2026).

All research and policy summaries are available at:

www.susannaberkouwer.com/research

Stove ownership 3.5 years later

- ▶ Among people who bought an improved stove from us:
 - ▶ 83% still own an improved stove
 - ▶ 34% still own a different wood/charcoal stove
- ▶ Among people who did not buy an improved stove from us:
 - ▶ 11% now own an improved stove
 - ▶ 88% still own a different wood/charcoal stove
- ▶ Enumerators physically inspected stove ownership

Important welfare gains for the poor

- ▶ The average buyer saves \$114 in fuel spending in the first year (\$284 over the stove's lifetime): often spent on food (53%), school fees, or other household items
- ▶ Each \$1 of subsidy generates \$17–49 in avoided climate damages
SCC of \$120–340 per tCO₂e (EPA, 2023); net of stove manufacturing cost
- ▶ Including fuel savings: each \$1 of subsidy generates \$26–58 in total societal welfare gain
- ▶ Health benefits:
 - ▶ Meaningful reduction in self-reported respiratory symptoms: cough, red eyes, sore throat, runny nose
 - ▶ No impact on clinical health or health expenditures
 - ▶ Evidence against measurement error and experimenter demand

Improved stoves reduce charcoal usage 27–48%

- The independent variable is *ownership*: the presence of rebound effects, low usage, has lowered these estimates from what they would otherwise be

	1 month later			1 year later		3.5 years later	
	(a) kg	(b) USD	(c) USD	(d) USD	(e) USD	(f) $\mu\text{g}/\text{m}^3$	(g) USD
Owns improved stove (=1)	-1.48 (0.27)	-2.28 (0.29)	-1.55 (0.34)	-2.50 (0.42)	-2.68 (0.47)	-45.30 (19.48)	-1.35 (1.00)
Observations	796	7853	911	6979	855	593	696
Control Mean	3.37	5.72	4.96	6.17	6.30	93.74	5.00
Treatment Effect	-0.44	-0.40	-0.31	-0.41	-0.42	-0.48	-0.27
SES controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Data Source	Buckets	SMSes	Survey	SMSes	Phone	PA-II	Survey